Practice workbook · Explain how airway features keep air clean and flowing

Nasal warming and filtering, cartilage rings, mucus and cilia, and what happens when bronchitis or smoke damages them.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-AIRWAY-PROTECTION-AIRFLOW

1. Question 6 · part b · 2 marks

June 2019 Paper 1

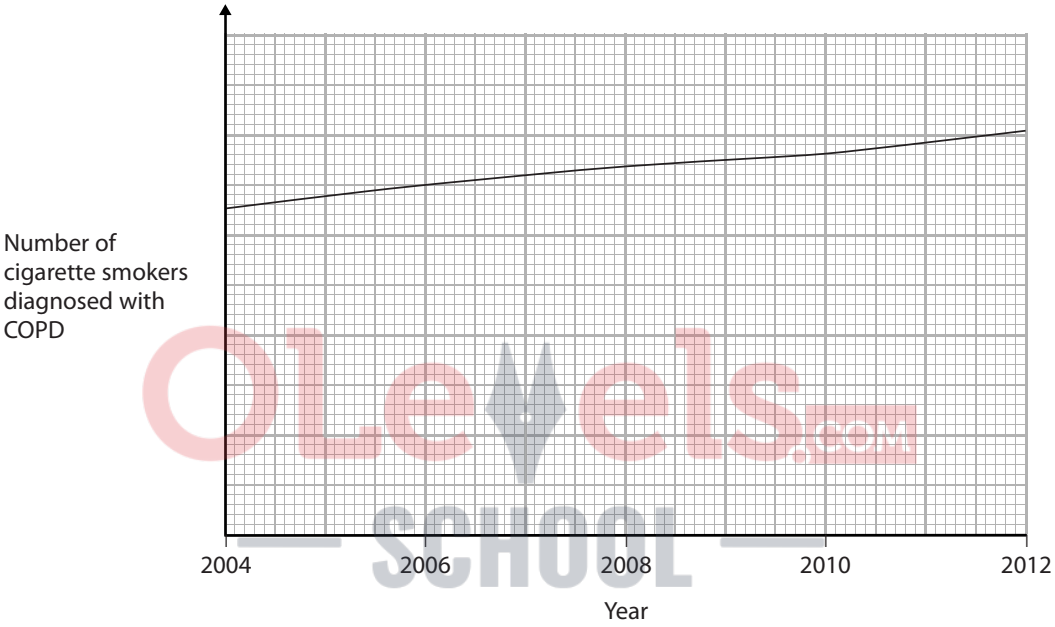
Worked-example retry · Attempt part b only.

6 (a) Chronic obstructive pulmonary disease (COPD) is the name given to a group of diseases that affect the breathing system.

These diseases include chronic bronchitis and emphysema.

Cigarette smoking is the main cause of COPD.

The graph shows the number of cigarette smokers diagnosed with COPD in the UK over a period of eight years.



(i) Describe the overall trend in cigarette smokers diagnosed with COPD.

(1)

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(ii) Describe what further information is required to help form the conclusion that cigarette smoking is the only cause of COPD.

(2)

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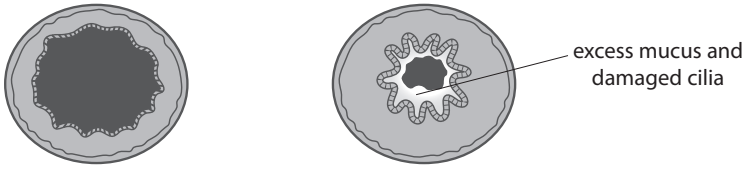
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1. Question 6 · part b · 2 marks

June 2019 Paper 1

Worked-example retry · Attempt part b only.

(b) The diagram shows how chronic bronchitis affects the airways in the breathing system.



healthy airway

chronic bronchitis

Explain how excess mucus and damaged cilia affect the breathing system of a person with chronic bronchitis.

(2)

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1. Question 6 · part b · 2 marks

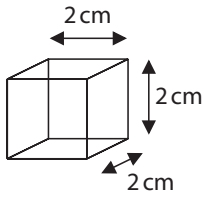
June 2019 Paper 1

Worked-example retry · Attempt part b only.

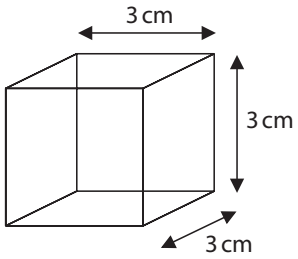
(c) The diagram shows models of two alveoli.

One model represents an alveolus from a person with healthy lungs.

The other model represents an alveolus from a person with emphysema.



Healthy



Emphysema

The table shows the surface area to volume ratio for a healthy alveolus.

	Surface area in cm^2	Volume in cm^3	Surface area to volume ratio
Healthy	24	8	3 : 1
Emphysema			

(i) Complete the table by giving the missing information.

(3)

(ii) Explain how the surface area to volume ratio of alveoli in the lungs of a person with emphysema will affect the normal function of body cells.

(3)

(Total for Question 6 = 11 marks)

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1. Question 6 · part b · 2 marks · Working and answer

Working and answer

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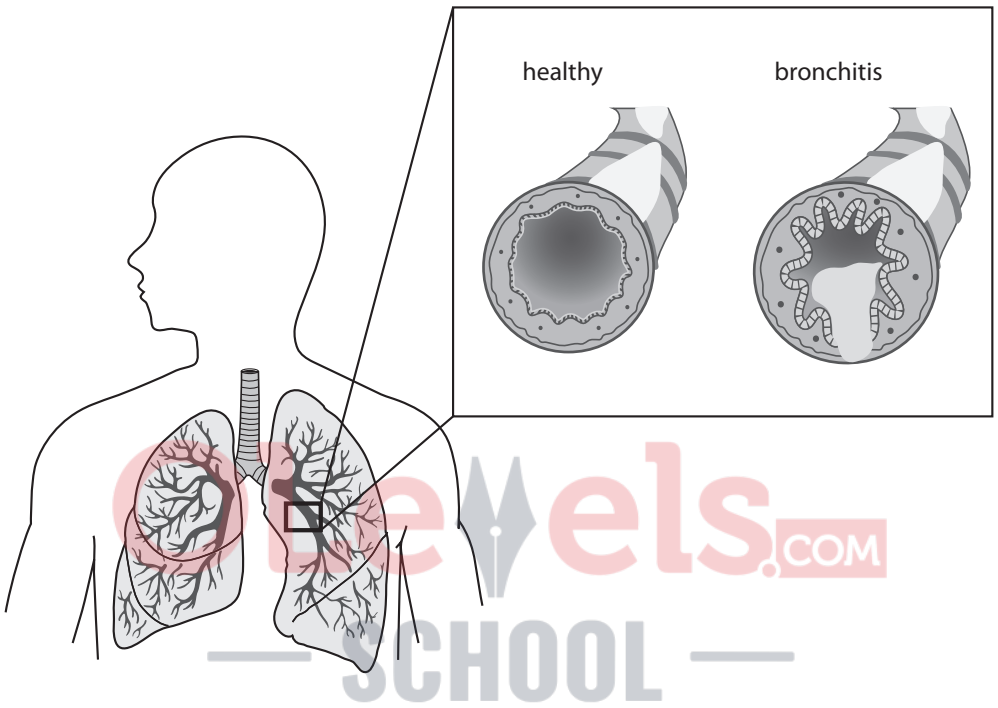
2. Question 6 · part b.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.iii only.

6 Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.



(a) Name the part of the lungs affected by bronchitis.

(1)

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2. Question 6 · part b.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.iii only.

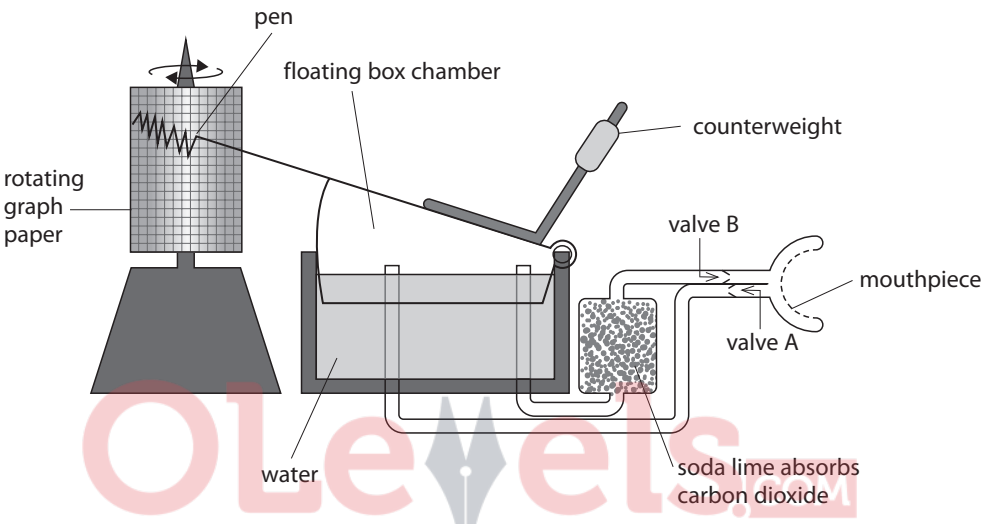
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(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.



(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

(4)

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2. Question 6 · part b.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.iii only.

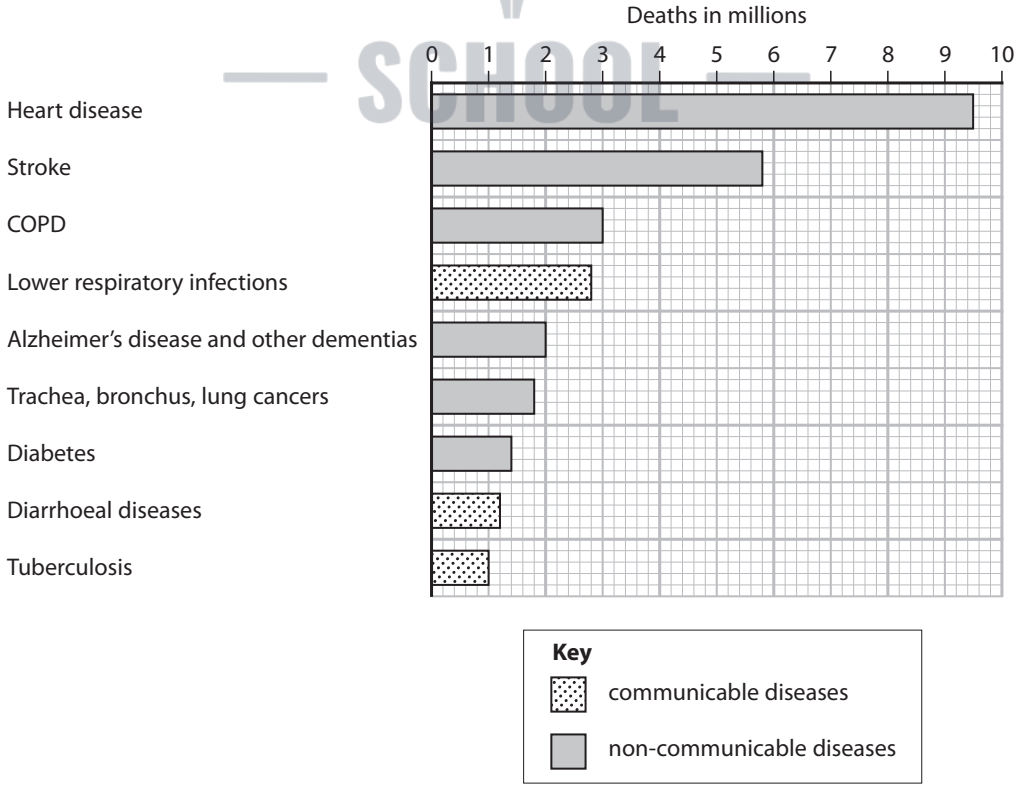
(ii) State one way to reduce risk to health when using a spirometer.

(1)

(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

(2)

(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.



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2. Question 6 · part b.iii · 2 marks · Working and answer

Working and answer

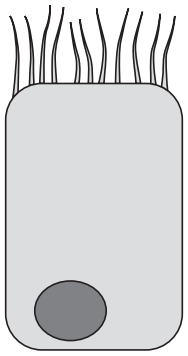
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3. Question 2 · part b.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b.ii only.

2 The diagram shows a cell from the human breathing system.



(a) (i) Add these labels to the diagram.

- cilia
- cell membrane
- nucleus

(3)

(ii) Each of the cell structures has a particular function.

The boxes give a list of structures and functions.

Draw one straight line from each structure to its function.

(3)

Structure	Function
cilia	holds the genetic code
cell membrane	site where most chemical reactions occur
nucleus	controls the entry and exit of substances
	makes proteins
	moves mucus through the trachea

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3. Question 2 · part b.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b.ii only.

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(b) (i) Cigarette smoking affects the function of the cell shown in the diagram.

Name a substance in cigarette smoke that causes damage to this cell.

(1)

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(ii) Describe how cigarette smoking affects the cell shown in the diagram.

(2)

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(Total for Question 2 = 9 marks)

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Working and answer

4. Question 2 · part a.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part a.i only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The breathing system consists of airways that allow the movement of air into the lungs. Air can be breathed in through the nose or through the mouth. The lining of the nose has a good supply of blood capillaries and a layer of sticky mucus.

The whole of the trachea and the larger air passages have C-shaped rings of cartilage within their walls. These rings help the air to flow freely into the lungs. The trachea divides into two tubes which then divide further into finer branches. These finer branches eventually end in air sacs. It is in the air sacs that gaseous exchange occurs.

The lungs are located in the thorax which is surrounded by the ribs. The thorax is separated from the abdomen by the diaphragm. Together, the ribs and the diaphragm can change the volume of the thorax and therefore the pressure in the thorax.

During breathing, at rest, an average of 0.50 dm^3 of air is breathed in and out. The air breathed in contains 20% oxygen and the air breathed out contains 16% oxygen. At rest, an adult will breathe 17 times a minute on average. The breathing rate is controlled by a part of the brain which receives information about carbon dioxide levels in the blood.

Carbon monoxide may be present in the air that is breathed in. The carbon monoxide can enter the blood and combine with haemoglobin in the red blood cells. Carbon monoxide is found in cigarette smoke and car exhaust fumes.

(a) (i) Discuss whether it is better to breathe in through the nose or the mouth. (lines 2 to 3)

(3)

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4. Question 2 · part a.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part a.i only.

(ii) Explain why the larger air passages have C-shaped rings of cartilage within their walls. (lines 4 to 5)

(2)

(iii) The trachea divides into two tubes.

What is the name of these tubes? (line 6)

(1)

- ☒ **A** alveoli
- ☒ **B** bronchi
- ☒ **C** bronchioles
- ☒ **D** oesophagus

(b) Which of these is the name of the part of the brain that controls breathing rate? (line 16)

(1)

- ☒ **A** cerebrum
- ☒ **B** cerebellum
- ☒ **C** hypothalamus
- ☒ **D** medulla oblongata

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4. Question 2 · part a.i · 3 marks · Working and answer

Working and answer

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5. Question 2 · part a.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.ii only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

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5. Question 2 · part a.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.ii only.

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(ii) Explain why the larger air passages have C-shaped rings of cartilage within their walls. (lines 4 to 5)

(2)

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5. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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